

VELMÈRE BASIC AUDIT REPORT

Stellar Consensus Protocol (native:stellar-core)

VELMERE SECURITY AUDIT REPORT: STELLAR CONSENSUS PROTOCOL

Audited Contract: native:stellar-core

Network: Stellar Network (Chain ID: stellar:pubnet)

Report ID: rep_xlm_basic_112 | Tier: BASIC | Application Surface: SHIELD

Created At: 2026-09-10T12:36:17.272Z

VERDICT SUMMARY:

ELEVATED RISK (UNVERIFIED CODE) (Risk: 72/100 | Quality: 64/100)

Release Decision: BLOCKED | Verification Status: INSUFFICIENT_EVIDENCE

STOP-SELL DIRECTIVE: INSUFFICIENT EVIDENCE: Coverage below minimum threshold for institutional sale

Confidence Score:

60/100

Evidence Coverage:

20%

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

[VERIFIED - IMMUTABLE]

Root: sha256:f44e23fc5c488b25...dbc8b3ea5989



Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

L1 Network Resilience Matrix: Consensus Verification 0% | P2P Topology Graph 0% | Node Fault Detectors 100% | Cryptographic Verification 0%

Contract lacks verified on-chain runtime bytecode. In accordance with Velmère evidence standards, missing bytecode results in an elevated risk assessment (72/100).

PROTOCOL OVERVIEW & NATIVE L1 NETWORK CONTEXT

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of distributed architecture and consensus mechanism

Protocol / Asset: Stellar Consensus Protocol

Architecture Type: Native Base Layer (Layer-1 · No Smart Contract)

Network Identifier: native_chain:xlm

Native Symbol: XLM

Rule Governance: Distributed node consensus across validators / miners

CONSENSUS SPECIFICATION & NODE CLIENT VERIFICATION

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Analysis of block validation rules, transactions, and cryptographic curves

Verification of Stellar Consensus Protocol is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT_APPLICABLE for native base protocol assets.

BASELINE NETWORK SECURITY FINDINGS

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified risk vectors in base layer architecture

CONSENSUS GOVERNANCE & PROTOCOL UPGRADE ANALYSIS (PRO)

PRO

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Distributed consensus governance and protocol upgrade mechanism analysis
- Validator / miner distribution and network finality threshold evaluation
- Hard-fork coordination history and client implementation diversity audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

SPOT MARKET DEPTH, EXCHANGE RESERVES & NETWORK VELOCITY (PRO)

PRO

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Global order book depth across regulated spot venues and custodial reserves
- On-chain circulating supply velocity and active address distribution
- Network transaction fee volatility and mempool congestion behavior
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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Document integrity verified by Velmère | Ref rep_xlm_basic_112 / bf7fc59087c4d6d5

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NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO)	PRO
[LOCKED]	[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Consensus reorganization and majority hashrate/stake attack modeling
- P2P network partition, eclipse attack, and sybil resistance analysis
- Mempool front-running, transaction censorship, and MEV extraction vectors
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED)	ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Reference node client reproducible build verification against repository tags
- Consensus serialization and wire protocol binary compatibility analysis
- State transition engine deterministic execution boundary checks
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED)	ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Protocol upgrade proposal activation and deployment history
- Backward compatibility verification of block validation logic
- Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED)	ADVANCED
[LOCKED]	[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed analyst attestation digest
- Manual review boundary explicit: no guarantee of absolute perfection
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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